

Professional Summary

Research scientist in generative modeling with a background in theoretical and computational physics. Since 2022, developing diffusion, flow-matching, and normalizing-flow models that respect exact symmetries, for sampling quantum field theories on grids, together with open-source PyTorch software that supports distributed training. Earlier work in precision lattice QCD and mathematical physics brings the mathematical depth to design models and make them reliable.

Research impact: 27 peer-reviewed publications ([full list](#)), h-index 20, 3426 citations ([Google Scholar](#), Oct 2026).

Research Highlights

- **Parameter-efficient generative architectures** for physics simulations, where Monte Carlo slows down near critical points, building in exact symmetries:
 - PSD flow: a Fourier-space layer inspired by effective field theory that reshapes correlations at all scales at once, in a model with only $\sim 3.4\text{K}$ parameters.
 - SVD-based gauge-equivariant flows: a 1288-parameter block on a 4^4 SU(3) lattice achieves effective sample size comparable to a 288-layer spectral-flow baseline.
- **Symmetry-aware diffusion models:**
 - Equivariant diffusion: reverse-time SDEs from an exact-likelihood ODE to Langevin-corrected sampling; much shorter autocorrelation than Hybrid Monte Carlo (HMC).
 - Diffusion on Lie groups: implicit score matching; the SDE family extended to Lie groups; compact gauge-equivariant networks and a multi-scale U-Net.
 - Noise schedule on Lie groups: derived analytically; gives linear energy decay and few-step sampling without the hand-designed drift Euclidean models need.
- **Machine learning combined with statistics:** symmetries split a sampling problem into independent pieces learned by a tiny flow, plus an exact statistical reweighting that removes the bias; validated against HMC.
- **Precision physics:** High-precision determinations of Standard Model parameters (quark masses, decay constants) from large-scale lattice simulations in international collaborations, using effective field theories; results in the particle-physics world averages (FLAG, PDG).
- **Mathematical physics and hyperasymptotics:** Resummation of divergent perturbative series, including the minimal renormalon-subtracted (MRS) scheme used for precise heavy-quark masses; methods for nonlinear eigenvalue problems.

Research & Technical Experience

2021–2026 Postdoctoral Researcher, *ETH Zurich*

- Developed generative-model methods, open-source software, and distributed-training infrastructure.
- Integrated normalizing-flow modules into the EU [interTwin Digital Twin Engine](#).
- Co-supervised PhD, MSc, BSc, and intern projects in computational physics and scientific ML.

2015–2021 Postdoctoral Researcher, *TU Munich / Glasgow / Tehran*

- Monte Carlo simulations and Bayesian parameter estimation for precision lattice-QCD observables.
- Developed scalable lattice-QCD simulation pipelines on HPC systems in international collaborations.

Open-Source Scientific ML Software

`lattice_ml` **Composable PyTorch library for lattice field theory (v0.2.0)**

- Adjoint-method ODE integrators for memory-efficient backpropagation of derivatives through long trajectories, compatible with automatic differentiation, for scalar and Lie-group-valued objects; integrators that stay on the group manifold; symplectic solvers; Hutchinson trace estimation.
- Implicit score matching and diffusion on Lie groups with predictor–corrector sampling; flow matching and flow maps.
- HMC for $U(1)$ and $SU(N)$; AD-safe SVD and eigendecomposition; gauge-equivariant layers; prelink and holonomy tools; Lie-group coordinates and random-matrix sampling; distributed training.
- Used in arXiv:2605.06134, 2605.17326, and 2610.09147 ([reproducible notebooks](#)).

`normflow` **PyTorch package for normalizing flows in lattice field theory (v3.0.0, since 2021)**

- Scalar and $SU(N)$ gauge theories; self-learning from the action (reverse KL with path gradients) or data-based training; asymptotically exact sampling via a Metropolis step.
- EFT-inspired PSD and Fourier-space flows; coupling, rational-quadratic spline, and autoregressive layers.
- Spectral and SVD-based gauge-equivariant flows; distributed training.

Technical Skills

Programming	Python, C++, Cython, Bash; Linux, Git
ML/AI	PyTorch, Diffusion Models, Flow Matching, Normalizing Flows, Score Matching, Equivariant Deep Learning, Efficient Sampling (HMC, Predictor–Corrector), Distributed Training
Mathematics	Differential Equations, Linear Algebra, Lie Groups, Asymptotic Analysis, Optimization
Statistics	Monte Carlo Methods, Bayesian Inference, Importance Sampling, Statistical Modeling
Computing	CUDA, MPI, HPC and Supercomputing Environments, NumPy, SciPy
Languages	English (fluent), German (basic, A2)

Mentoring & Leadership

Mentoring **Co-supervised PhD, MSc, BSc, and intern researchers in physics & scientific ML:**

- PhD (2): gauge-equivariant diffusion models; Fourier acceleration for $SU(N)$.
- MSc (6): equivariant U-Net; autoregressive, continuous-time, and spline-based flows; simulated tempering.
- BSc and intern: path-integral Monte Carlo; CNNs in arbitrary dimensions.

Organization **Co-organized workshops on physics & machine learning:**

- ZPW 2026 — Machine Learning Techniques for Particle Physics.
- QCD 2025 — Symposium on QCD Factorization.
- ML Lattice 2025 — Machine learning approaches in Lattice QCD.
- UT 2019 — Lattice QCD workshop; TUM 2017 — EFT and Lattice Field Theory Summer School.

Refereeing Referee for Physical Review D (13+ reviews since 2016).

Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
Advisors: Claude Bernard and Carl M. Bender

- Developed EFT-based frameworks and Bayesian inference methods for lattice QCD.
- Performed precision determinations of hadronic observables in lattice QCD.
- Developed a novel definition of eigenvalue problems for nonlinear systems arising in physics.

2009 **MSc in Electrical Engineering**, *University of Tehran*

- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*